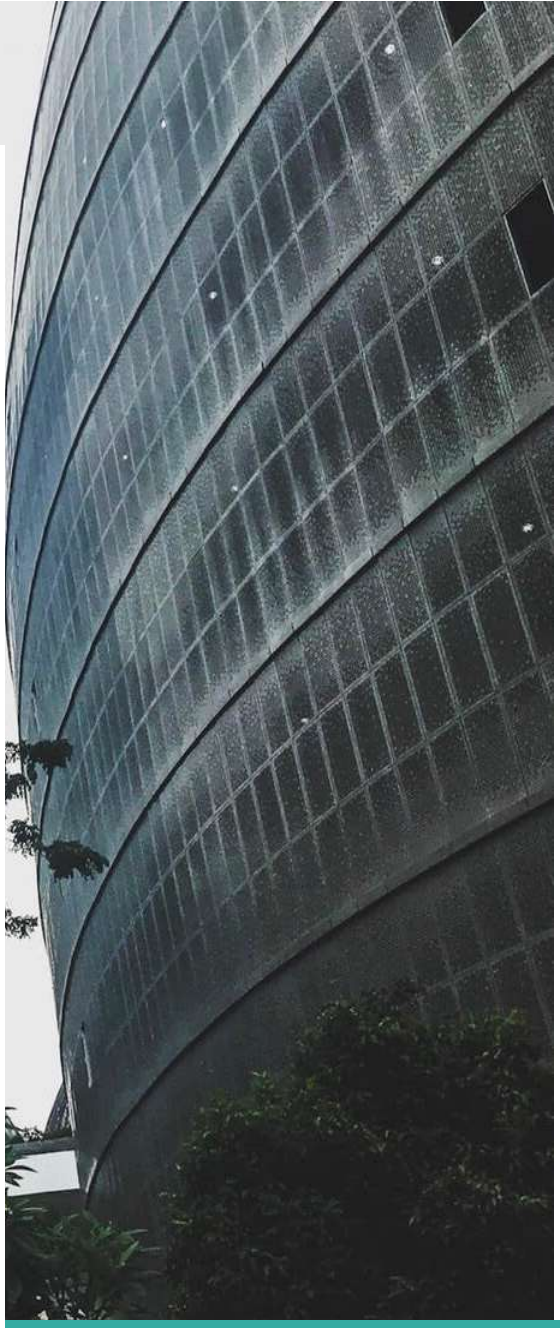
MODUL PEMBELAJARAN

IF670 – Mobile Cross Platform Programming
S1 INFORMATICS GRADUATE PROGRAM
FACULTY OF ENGINEERING AND INFORMATICS



PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
UNIVERSITAS MULTIMEDIA NUSANTARA

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MODUL 9
(Camera, Storage, Filesystem)

DESKRIPSI TEMA

Mahasiswa akan mempelajari cara menghandle camera, storage, dan filesystem.

CAPAIAN PEMBELAJARAN MINGGUAN (SUB-CAPAIAN PEMBELAJARAN)

Mahasiswa mampu memanfaatkan fitur native seperti kamera, penyimpanan, dan sistem file dalam pengembangan aplikasi.

PENUNJANG PRAKTIKUM

1. Laptop / PC
2. IDE (Android Studio/XCode)
3. Code Editor (Visual Studio Code/Sublime/etc)

LANGKAH-LANGKAH PRAKTIKUM

1. Buatlah sebuah projek baru dan install beberapa library pendukung:
npx expo install expo-camera expo-media-library expo-image-picker

2. Tambahkan code berikut pada dalam expo di app.json.

```
"plugins": [
  "expo-router",
  [
    "expo-splash-screen",
    {
      "image": "./assets/images/splash-icon.png",
      "imageWidth": 200,
      "resizeMode": "contain",
      "backgroundColor": "#ffffff",
      "dark": {
        "backgroundColor": "#000000"
      }
    }
  ],
  [
    "expo-camera",
    {
      "cameraPermission": "Allow $(PRODUCT_NAME) to access your camera",
    }
  ],
  [
    "expo-media-library",
    {
      "photosPermission": "Allow $(PRODUCT_NAME) to access your photos",
      "savePhotosPermission": "Allow $(PRODUCT_NAME) to save photos to your library",
      "isAccessMediaLocationEnabled": true
    }
  ]
],
```

3. Modifikasi index.tsx anda untuk menggukan kamera dan mengakses gallery.

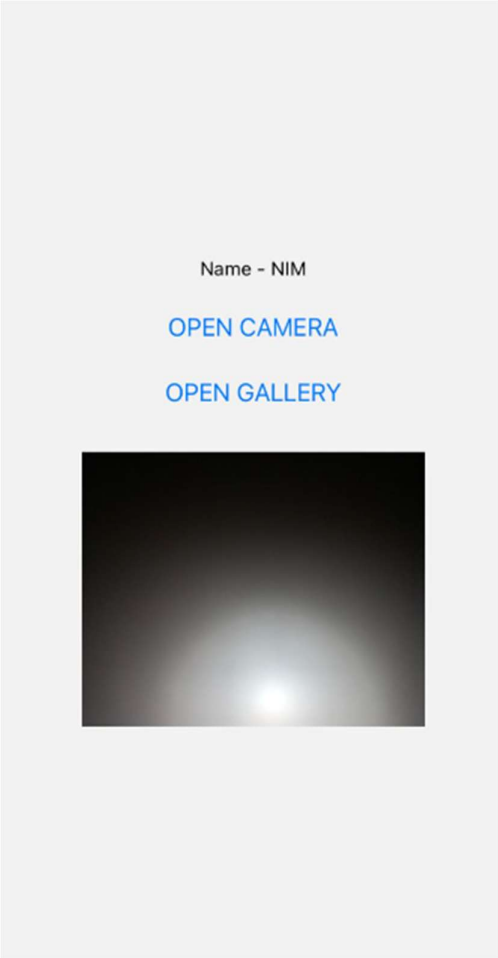
```
import { Text, View, Button, Image, StyleSheet } from "react-native";
import { useState } from "react";
import * as ImagePicker from "expo-image-picker";
import { Camera } from "expo-camera";

export default function Index() {
  const [image, setImage] = useState<string | null>(null);

  // 📷 OPEN CAMERA
  const openCamera = async () => {
    const permission = await Camera.requestCameraPermissionsAsync();

    if (!permission.granted) {
      alert("Camera permission is required!");
      return;
    }

    const result = await ImagePicker.launchCameraAsync({
      mediaTypes: ImagePicker.MediaTypeOptions.Images,
      quality: 1,
    });

    if (!result.canceled) {
      setImage(result.assets[0].uri);
    }
  };
```

```
// 🖼 OPEN GALLERY
const openGallery = async () => {
  const permission = await ImagePicker.requestMediaLibraryPermissionsAsync();

  if (!permission.granted) {
    alert("Gallery permission is required!");
    return;
  }

  const result = await ImagePicker.launchImageLibraryAsync({
    mediaTypes: ImagePicker.MediaTypeOptions.Images,
    quality: 1,
  });

  if (!result.canceled) {
    setImage(result.assets[0].uri);
  }
};
```

```
  return (
    <View style={styles.container}>
      <Text style={styles.text}>
        Name - NIM
      </Text>

      <View style={styles.button}>
        <Button title="OPEN CAMERA" onPress={openCamera} />
      </View>

      <View style={styles.button}>
        <Button title="OPEN GALLERY" onPress={openGallery} />
      </View>

      {image && (
        <Image source={{ uri: image }} style={styles.image} />
      )}
    </View>
  );
}
```

```
const styles = StyleSheet.create({
  container: {
    flex: 1,
    alignItems: "center",
    justifyContent: "center",
  },
  text: {
    marginBottom: 10,
  },
  button: {
    marginVertical: 5,
    width: 150,
  },
  image: {
    width: 250,
    height: 200,
    marginTop: 20,
  },
});
```

4. Berikut adalah hasil implementasi kode diatas



6. Setelah anda berhasil mengimplementasikan kamera, install library pendukung:
npx expo install expo-file-system

TUGAS

Modifikasilah tutorial yang anda telah kerjakan. Buatlah fungsi dan button "SAVE IMAGE" dengan memakai library expo-file-system yang sudah di-install sebelumnya, lalu cek apakah benar sudah tersimpan di gallery anda.

REFERENSI